SafePath

An SMS-first mobility companion for women in Tamil Nadu
Ideathon Brief v2 (refocused): SDG 5 Gender Equality

One line: A woman with a basic button phone texts in Tamil. She gets the **free bus** for her trip, a **safe place to wait** run by local women's Self-Help Groups, and an automatic **Journey Watch** that texts her family if she doesn't reach on time. Every request (anonymised) also feeds a **dashboard** that shows the transport department where women's travel needs are not being met.

PROJECT OVERVIEW

What we're building	An SMS-first assistant (no smartphone, no internet, no app needed), run by 4 AI agents connected through n8n, plus a web dashboard for policy makers.
Who it's for	Women (and transgender persons) in Tamil Nadu who travel by bus, especially those with basic phones, low data, or limited English.
SDG 5 targets	**5.2** Ending violence against women in public spaces · **5.5** Women's economic participation · **5.b** Technology that empowers women
What's different	Works on a button phone · prevents problems instead of only reacting to SOS · built on Tamil Nadu's existing free bus scheme · uses the existing SHG network of women
Timeline / team	72 hours · 4 people
Stack	n8n · Gemini / Claude · Sarvam AI (Tamil) · Supabase · SMS gateway · Vercel

THE PROBLEM: MOBILITY IS A GENDER GAP

This is the answer to the judges' first question: *"Why is this gender equality and not just a better bus app?"*

- **Women travel differently.** They make more short, chained trips (school, market, work, back home), travel more off-peak, and depend on buses more than men do.
- **Fear limits opportunity.** When a woman (or her family) thinks a trip is unsafe, she turns down jobs, colleges and training that are too far or end too late. Unsafe transport becomes lost income and lost education.
- **Tamil Nadu already made bus travel free for women** (the *Vidiyal Payanam* scheme, on ordinary town buses). But many women can't easily tell which buses on their route are free, and the information mostly lives in smartphone apps.
- **Most safety tools need a smartphone** and only act *after* something has gone wrong (the SOS button). The women with the least access are left out.

[VERIFY before slides] Find 2–3 recent, sourced numbers: change in women's share of town-bus ridership since the free-bus scheme; share of women in TN using basic phones; one survey on women avoiding trips or jobs due to safety. Cite the source on the slide. Do not use numbers you can't source.

WHY EXISTING TOOLS FALL SHORT

Tool	What it does well	The gap SafePath fills
Kavalan SOS (TN Police)	Fast SOS to police	Smartphone only; acts after danger, not before
112 India / 181 Women Helpline	Emergency response	Reactive; no trip planning or family loop
Safetipin	Safety audits of public spaces	Smartphone app; no free-bus routing; no SMS
Chalo / transit apps	Live bus info	Not built around women's needs; smartphone only

Tool	What it does well	The gap SafePath fills
SafePath	**Free-bus route + safe waiting spot + family alert, all by SMS**	**Prevention, button-phone access, community network, policy data**

Position SafePath as **complementary**: in an emergency it points her to 112 / 181. It doesn't replace them.

THE 4 AGENTS (+ n8n ORCHESTRATION)

Each agent has one clear job and works for real in the demo. n8n receives every SMS, sends it to the right agent, and sends the reply back.

Agent	Job	Inputs → Output
1. Intake Agent	Understands a short, messy Tamil / English / Tanglish SMS (or a voice note via Sarvam) and works out the intent: route, start trip, reached, help, add contact.	Raw SMS → {intent, from, to, time, language}
2. Free-Bus Route Agent	Finds a route using **free (ordinary) buses** first, names the bus number and stop, and flags if only a paid bus runs at that time.	From / to / time → "Take 21G (FREE) from X stop, every ~15 min"
3. Journey Watch Agent	Watches the trip. If she doesn't reply REACHED in time, it checks with her first, then alerts her trusted contacts with her route and stop.	START + ETA → timers → SMS to contacts
4. Sakhi Network Agent	Suggests the nearest verified **safe waiting point** (an SHG-run stall or women-run shop) near her stop, with opening hours.	Stop + time → "Wait at Lakshmi Tea Stall, opp. stop, open till 9pm"

SMS commands (keep them dead simple)

She sends	What happens
Any normal message, e.g. *"T Nagar to Nungambakkam 8 mani"*	Intake works out the intent → Route + Sakhi reply in her language
START	Journey Watch starts a timer from the expected arrival time (ETA)
REACHED	Trip closed; optional "she reached safely" SMS to family
HELP	Immediate SMS to trusted contacts + nearest safe point + reminder to call 112 / 181
ADD 98xxxxxxxx	Adds a trusted contact (the contact gets an SMS and must reply YES to agree)
Missed call (stretch goal)	Callback with a Tamil voice menu (IVR) using Sarvam speech

Journey Watch: how it escalates

- **ETA + 10 min:** SMS to her: *"Reached? Reply REACHED or WAIT 15."* (Handles normal bus delays without panic.)
- **ETA + 25 min, no reply:** SMS to trusted contacts: *"Priya started from T Nagar at 8:05 on bus 21G and hasn't confirmed reaching. Please call her."*
- **Never** auto-dispatches police. False alarms would destroy trust. Humans she chose decide what to do next.
- Works with **no GPS**: it uses the route she planned plus timing. Location sharing can be an optional add-on for smartphones later.

THE POLICY DASHBOARD (THE SYSTEM-LEVEL IMPACT)

Every request is stored **without names or phone numbers**. Grouped together, it shows where women's needs are not being met:

- **Demand with no free bus:** "312 women asked for X → Y after 8pm this month; only paid buses run then."
- **Stops with no safe waiting point** after dark, where SHGs or the city could add one.
- **Journey Watch delays by route and time:** shows where women are waiting too long.

This moves SafePath from helping one woman at a time to **producing evidence that changes transport policy**. That's the SDG 5 story judges remember.

HOW IT WORKS: PRIYA'S DAY (THE DEMO STORY)

Priya, 28. Button phone, Tamil speaker, works a 9–6 job, travels with her 5-year-old daughter, nervous about waiting alone in the evening.

When	What happens
7:50 am	SMS: *"T Nagar to Nungambakkam, school 8:30, with paapa"* → Reply in Tamil: free bus number, stop, rough frequency, and a note that a paid bus is faster if she's running late.
8:02 am	She sends **START**. Journey Watch sets her ETA.
8:28 am	She sends **REACHED**. Her husband gets: "Priya reached safely."
8:40 pm	Return trip. Reply includes: *"Wait at Selvi's SHG tea stall opposite the stop (open till 9:30)."*
9:20 pm	No REACHED reply, and no reply to the check-in → her sister gets an SMS with her bus and boarding stop. Her sister calls her.
Dashboard	Her (anonymous) trip adds to the "evening demand, Nungambakkam stop" data that judges see on the web dashboard.

Honest scope: we save simple preferences (home stop, usual contacts, language). We do **not** claim "AI learns her routine" in 72 hours. Judges respect honest scope.

SAFETY & ETHICS PRINCIPLES (SAY THESE OUT LOUD)

- **No "danger scores" for neighbourhoods.** We only show positive signals: verified safe waiting points, open shops, free buses. Labelling areas "unsafe" without data stigmatises poorer areas and gives false confidence.
- **Trusted contacts only, never strangers.** No matching women with unknown "buddies". Safe points are vetted through SHG federations.
- **Consent both ways.** A contact must reply YES before receiving alerts. She can send STOP to delete her data.
- **Data minimisation.** The dashboard never stores phone numbers or names; trip logs auto-delete after a set period.
- **The family loop can cut both ways.** We note the risk of controlling relatives tracking women. That's why *she* starts each Journey Watch; it's never always on.

DATA SOURCES

Data	Source	For the 72h demo
Bus routes & stops (Chennai MTC)	Public route lists / open data portals / community GTFS datasets **[verify which exist]**	Hand-clean 5–10 real routes around the demo area into a table
Which buses are free	Vidiyal Payanam scheme: ordinary town buses (verify bus types)	A free / paid flag per route
Safe waiting points	SHG (Mahalir Thittam) stalls, women-run shops	15–20 **mock** points, clearly labelled as sample data; ideally 2–3 real ones verified in person
Live bus position	Not reliably open	**Skip.** Use timetables or typical frequency

TECH STACK

Layer	Choice	Note
Orchestration	n8n	One main workflow (SMS in → agent → SMS out) + one scheduled workflow for Journey Watch timers
LLM	Gemini (free tier) or Claude	Intake understanding + writing friendly replies. Route search is **plain code over the table**, not the LLM (no made-up bus numbers)

Layer	Choice	Note
Tamil language	Sarvam AI	Translation, plus speech-to-text / text-to-speech for the voice stretch goal
Database	Supabase	users, contacts, trips, routes, safe_points, anon_requests
SMS	Twilio trial *or* an Android phone used as an SMS gateway	Commercial Indian SMS needs DLT registration (takes days). Use a trial or a phone gateway for the demo
Dashboard	Simple web page on Vercel	Map + 3 charts; read-only from anon_requests

72-HOUR BUILD PLAN

Phase	Hours	What gets done
0. Lock scope	0–2	Freeze this brief. Pick the demo area (one corridor, e.g. T Nagar ↔ Nungambakkam). Set up accounts.
1. Quick research	2–14	Short 10-question form to **20–40 women** you can actually reach (college, family, neighbours) + 3–5 short phone interviews. Goal: 3 real quotes + 3 numbers for the slides, not a full study.
2. Data	2–14 (parallel)	Clean 5–10 routes, free/paid flags, 15–20 safe points into Supabase.
3. Core loop	14–36	SMS in → n8n → Intake → Route + Sakhi → SMS out, working end-to-end in Tamil and English.
4. Journey Watch	36–50	START / REACHED / HELP, timers, contact consent, escalation SMS.
5. Dashboard	36–54 (parallel)	Anonymous request log → map + charts.
6. Test with real users	50–60	5–8 women try it on their own phones. Fix the top 3 issues. Collect quotes.
7. Pitch	60–72	Record a backup demo video, 10 slides, rehearse the Q&A twice.

Stretch goals, only if the core loop is solid by hour 50: missed-call voice menu with Sarvam; WhatsApp channel; Tamil voice-note input.

TEAM (4 PEOPLE)

Person	Main role	Also owns
1	n8n workflows + SMS gateway	Live demo on stage
2	Agents: prompts, Intake logic, Sarvam	Tamil reply quality
3	Data (routes, safe points) + Supabase + dashboard	Data-source slide
4	Research, user testing, ethics	Slides, narrative, backup video

PITCH: 10 SLIDES, 3-MINUTE DEMO

- **1.** Title + one line
- **2.** Problem: mobility is a gender gap (sourced numbers + one real quote)
- **3.** The overlooked user: button phone, Tamil, a free bus she can't easily look up
- **4.** Why existing tools miss her (comparison table)
- **5.** SafePath: 4 agents, one SMS
- **6. Live demo**: Priya's morning and evening (real phone on stage)
- **7.** Sakhi Network: SHG women as the safety network
- **8.** Policy dashboard: from one trip to policy evidence
- **9.** Safety & ethics principles + what we tested with real women
- **10.** Impact & scale: other TN cities, other free-bus states, a partnership with the transport department

JUDGE Q&A: PREPARE THESE ANSWERS

Likely question	Your answer (short)
"How is this gender equality, not just transport?"	Mobility decides access to jobs and education; fear of travel costs women opportunities. We tackle access (free bus), safety (5.2), and give policy makers evidence.
"Why not just use Kavalan / 112?"	Those are for emergencies, on smartphones. We prevent problems, work on a button phone, and send her to 112 / 181 when needed.

Likely question	Your answer (short)
"Who verifies safe points?"	SHG federations / ward-level women's groups. Demo points are sample data, and we say so.
"What if family uses it to control her?"	She starts each watch; it's never always-on tracking; STOP deletes everything.
"What does it cost to run?"	SMS is the main cost (paise per message). Fits a CSR or government-scheme pilot. [Get a real per-SMS price before the pitch.]

WHAT WE DELIBERATELY CUT FROM v1 (AND WHY)

- **7 agents → 4.** Prediction / Learning agents can't genuinely learn in 72h; faking it costs credibility.
- **Safety scoring of areas.** Replaced with positive, verified safe points (see ethics).
- **4 channels → SMS first.** One channel that works beats four half-built ones.
- **75-woman survey in 4 hours.** Replaced with a smaller, real sample + honest user testing.

"Every woman should be able to travel free, safely and confidently, even from a Rs. 1,000 phone."